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ENC5705 Composition Theory and Pedagogy
3 December 2025

The Intersection Of Tech & The Writing Process, An Autoethnography

Introduction

While we often assign students assignments that helps them reflect on their writing processes, we rarely spend enough time doing the same with our own writing. This introspective look into my own writing process shows an example of how intertwined the writing process is with chosen and available technology and how a writing process can be very non-linear, while also serving as an example students seeking to analyze their own writing process can look to. This study shows how writing is a recursive rather than linear process, a process that creates many small cycles of thinking, writing, & revising to eventually build up to a final work.

Methods

This study reviews the writing process for an assignment in the genre called a ‘reflective synthesis’. The purpose of the assignment was to “(1) use writing as a way of analyzing relationships among the readings, (2) offer a useful heuristic for reflection, and (3) provide a culminating experience for a given set of readings and a given topic.” The parameters of the assignment were roughly 1 page in length, single-spaced, and a large amount of freedom in what topics to cover. There was no additional planning or brainstorming before sitting down to write. The software that I used in the assignment include Microsoft Word for word processing, Adobe Acrobat for reading and annotating PDFs, Zotero for citation management, and Google Chrome as an internet browser. While I will discuss hardware more in depth later, all the writing was completing on my home hobbyist PC.

The writing process was recorded using Open Broadcaster Studio (OBS), a free and open source software for video recording and streaming (*Open Broadcaster Software® | OBS*). Once recorded, I watched the video and took coded notes in Excel much like I would when observing over-the-shoulder. Task codes were Other, Writing, Revising, Editing, Rereading, Reading Other, and Navigating, while application codes were divided into Word, Acrobat, Zotero, Chrome, and OtherApp. These individually marked codes are combined to create a list of discrete codes represented as a series of Xs and Os, which was used to sort and analyze the data. The list of these codes and the sum of their count and time are available below. Finally, the timelines used for further analysis were generated using Visual Basic for Applications (VBA).

Code:	Action	Count	Total Time	Time By App	
OXOOOOOOXOOOO	WRITING WORD	50	0:23:37	0:42:58	WORD
OOXOOOOOXOOOO	REVISION WORD	11	0:09:45		
OOOXOOOOXOOOO	EDITING WORD	17	0:05:27		
OOOOXOOOXOOOO	READING WORD	12	0:03:56		
OOOOOOXOXOOOO	NAVIGATING WORD	2	0:00:13		
OOOOOXOOOXOOO	READING PDF	23	0:06:58	0:18:48	Acrobat
OOOOOOXOOXOOO	NAVIGATING PDF	21	0:06:02		
OOOOOXXOOXOOO	COMPLEX PDF	15	0:05:48		
OXOOOOOOXOXOO	ADD CITATION	10	0:01:14	0:04:01	Zotero
OOOOOOXOOOXOO	ZOTERO REFERENCE	4	0:01:12		
XOOOOOOOOOXOO	ZOTERO FUNCTION	2	0:01:35		
XOOOOOOOXOOOO	OTHER/BREAK	4	0:00:37	0:13:24	Other
XOOOOOOOOOOOX	OTHER/BREAK	10	0:09:42		
XOOOOOOOXOOXO	END	1	0:00:45		
OOOOOXXOOOOXO	BROWSER INTERACTION	4	0:02:20		

Figure 1- Chart with list of discrete tasks, their codes, and the final tally of their count and time spent.

Technology Analysis

The most immediate realization from reviewing the recording and coded data is how integrated specific technologies are in my writing process. I've owned a 32:9 resolution monitor since late-2019, and it's almost alarming how reliant I am on specific workflows enabled by this. Using a program called WinDock, I am quickly able to dock windows to different shapes and locations by dragging them to either the edges, corners, or prespecified areas, such as the bottom 20% of the screen. This took some time to initially adapt to, and there are certain applications that very much dislike being in a non-standard resolution, but it is now an important part of my home workflow. Figure 2 below shows the layout that my desktop was organized in during the observation period, and windows that I used were as follows:

- **Primary Working Window:** Microsoft Word (text processor)
- **Primary Reference Window:** Adobe Acrobat (PDF Viewer)
- **Auxiliary Window 1:** Zotero (Citation Manager)
- **Auxiliary Window 2:** Google Chrome (Internet Browser)
- **Flex Window 1:** Spotify (Music player)
- **Flex Window 2:** Facebook Messenger (Social media)

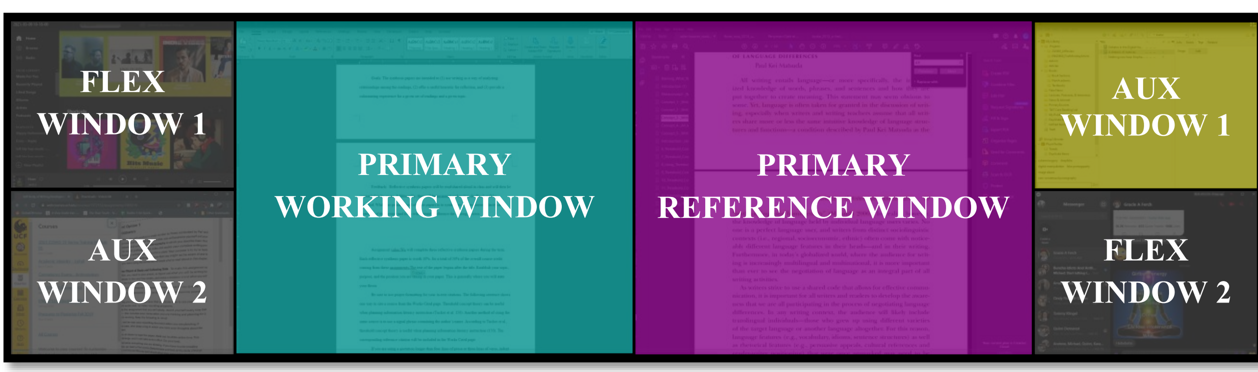

Figure 2- Diagram of desktop layout used during the writing process.

While Figure 2 shows the most common setup I use for writing in academic contexts, there are a number of different docking positions for windows set up. For example, I commonly combine the two primary windows into one for entertainment purposes, and Figure 3 shows an example of other window setups that are possible in different combinations:

Figure 3- Alternate desktop layouts.

While at no point during the observed period did I change my layout, these layouts and labels are rarely set in stone. At the time of writing this sentence, I'm using what I labeled as the "Primary Reference Window" for writing in Microsoft Word. But later on, when I would need to open acrobat to reference other authors, my layout gradually moved back over to the setup that I recorded. The large amount of screen real estate and the ability to move windows around quickly creates an environment that more readily mimics a physical working environment.

Comparing Time by Application

At 54%, just over half the time observed was spent with word as the active application. These periods include all the tasks highlighted in cyan in Figure 1: Writing, revising, editing, rereading what I had written, and navigating the Word application. Adobe Acrobat used 24% of the total time spent writing, usually reading or scanning through soft copies of sources I have stored on Zotero. While Zotero only encompassed about 5% of the total time spent, the time spent using the application almost certainly saved a magnitude or more in time spent managing sources and citations. Measuring time efficiency of these programs is outside the scope of this initial research project, but there are likely equally valuable time-savings found in the use of Word and Adobe over paper and hard copies of articles.

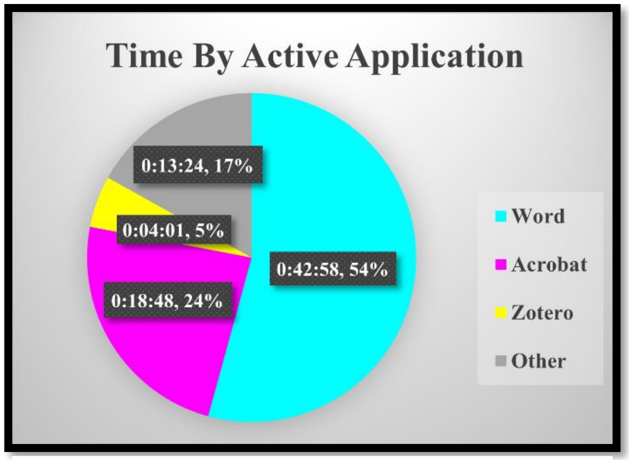

Figure 4- Time Spent by Active Application

Looking at the writing process as a timeline in Figure 5, it is easy to see how the writing process observed goes back and forth between using different software. Periods of writing were largely multi-modal, quickly going back and forth between a referenced document in magenta and the Word application in cyan. On the other hand, editing is seen primarily in the thick cyan blocks while changes to the Zotero database are in Yellow.

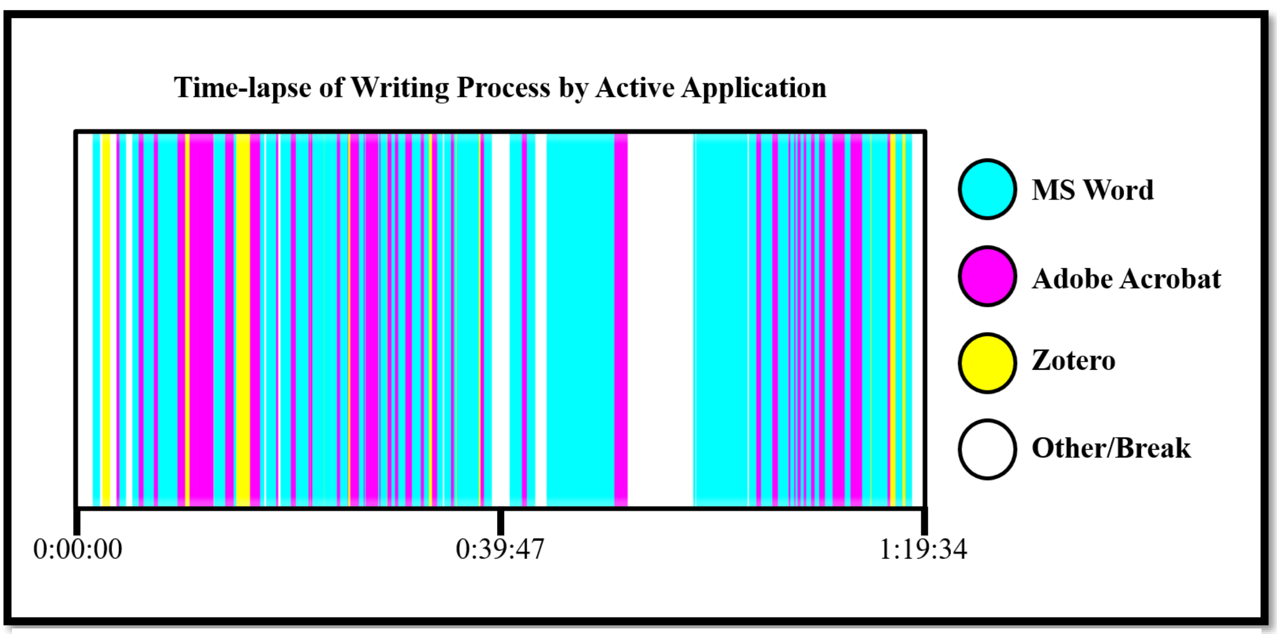

Figure 5- A timeline showing active application over time during the observed period.

Comparing Time by Time Spent Writing

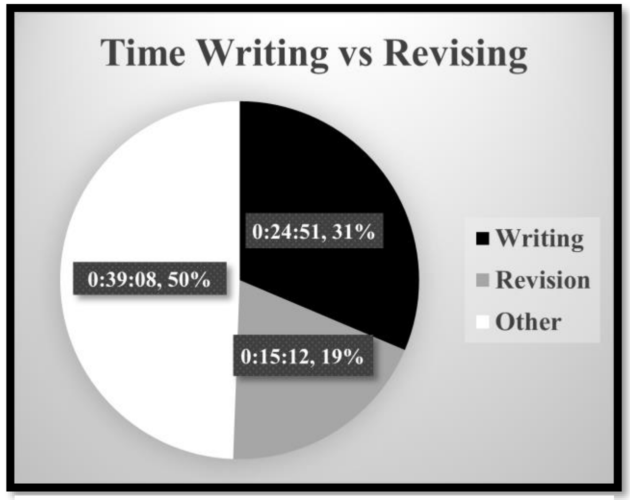

Figure 6- Time Spent by Active Application

Common writing advice found like on the Purdue Writing Lab often refers to the writing process in specific stages, such as **Brainstorming**, **Outlining, Rough Draft**, **Revise and Edit your writings**, and **Polish your writing** (Purdue Writing Lab). This process is written and often interpreted in metaphor as an assembly line, where each step must be completed before the next. In the writing I observed, the process was more similar to that of a constant back and forth between additive and subtractive manufacturing, respectively 3D printing compared to a lathe or mill. Figure # shows how quickly writing oscillated between the additive process of writing new material, the usually subtractive process of editing and revising, and periods of brainstorming or research. It looks like a jumbled, disorganized mess because, well, it is. Dedicated writing sessions are interrupted by serendipitous moments of revising to draw new connections or periods of research to find authors or specific passages that can support a point being made. The process is recursive in the way as described by Sommers that isn't always found in student writers (Sommers).

During that time period observed, only half the time was spent writing, revising, and editing, while the rest of time was used either researching, rereading what I had written, interacting with other software, and spending time away from the desk for short breaks. In looking at editing process, Sondra Perl draws the following conclusion: "Editing is primarily an exercise in error-hunting. The students are prematurely concerned with the "look" of their writing; thus, as soon as a few words are written on the paper, detection and correction of errors replaces writing and revising. Even when they begin writing with a tentative, flexible frame of mind, they soon become locked into whatever is on the page." (Perl). This pattern is found in my own writing process, where I quickly interrupt writing to make quick edits. However, the tools I used made making these edits so quick that they rarely completely interrupted full trains of thought. I would leave a sentence unfinished to spot-correct a minor error and, within a few strokes or clicks, go right back into my main point. Most of the time spent in the revision code was spent reorganizing ideas or building more support to specific arguments.

One final note to make about the time spent in each activity is that this was a school assignment rather than a piece intended for publication. Because of the lower stakes and competing time priorities, I did not feel the need to do a final copyediting pass on the document to do any more than a rough polish that happened during the writing process. Looking closely at the timeline, the very last writing activity was a period of fresh writing without any following edits. In retrospect, I would've liked to have completed this step just to have the data, but this final polish would usually be part of the writing process.

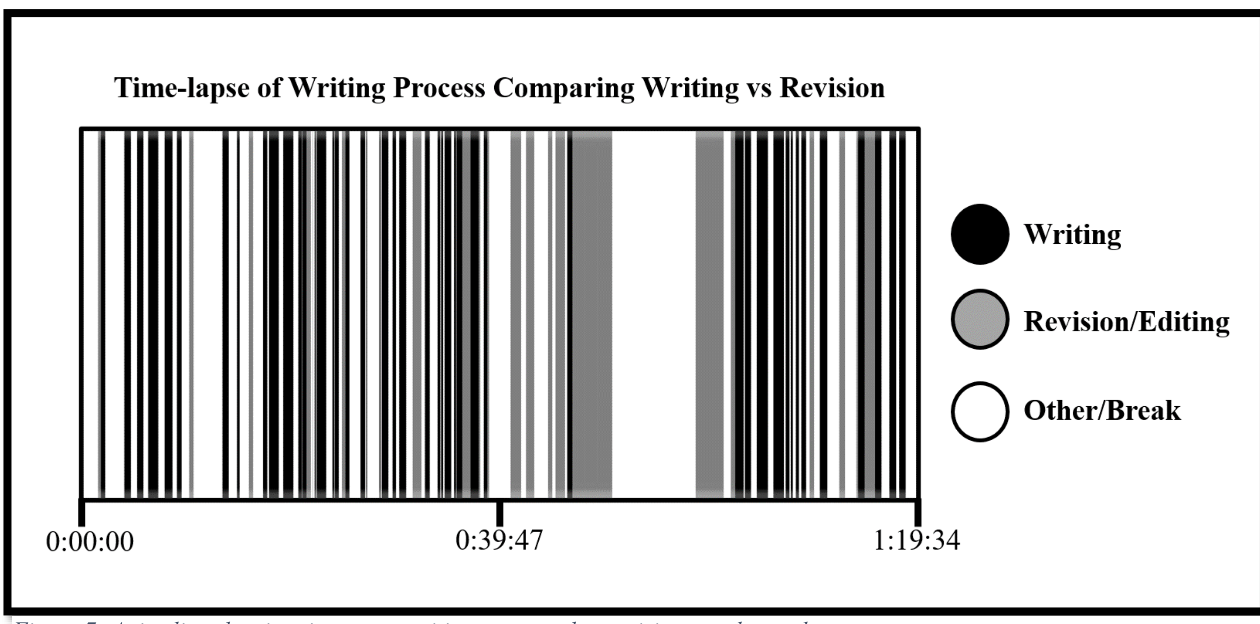

Figure 7- A timeline showing time spent writing compared to revision or other tasks..

Conclusion

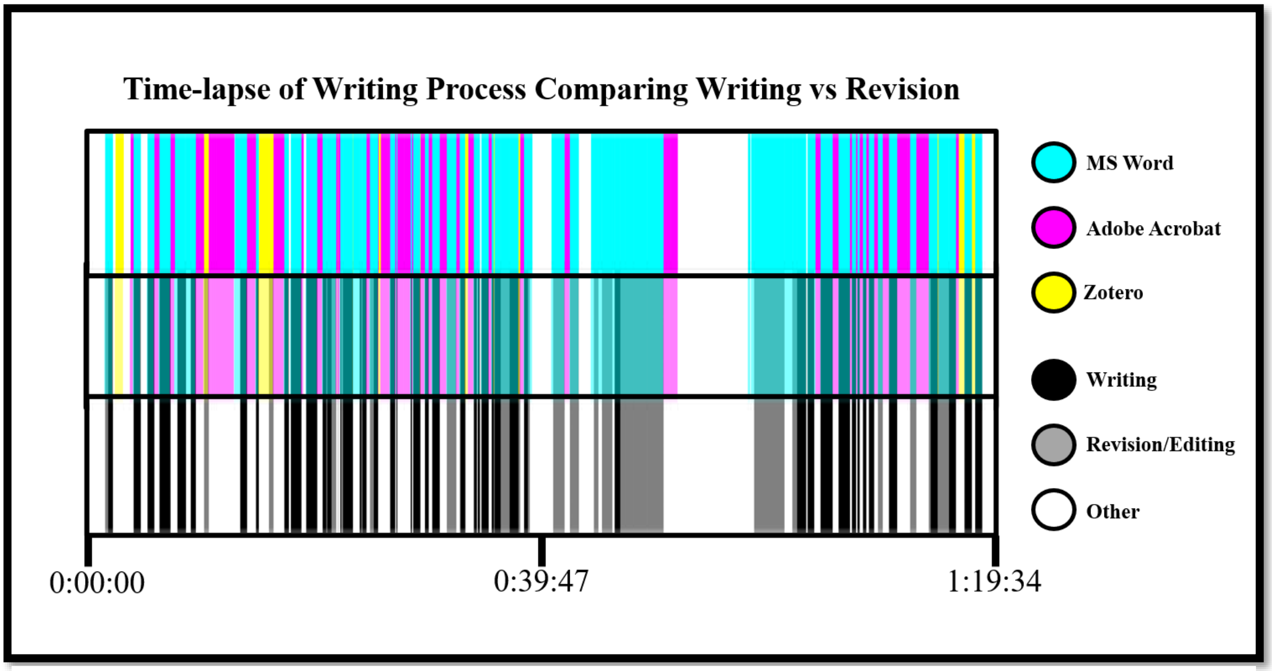

Figure 8- A combination of both timelines.

Looking at both timelines together, it is possible to see that different writing tasks manifested in different patterns of software usage. Large, interrupted blocks of cyan are indicative of revision, while new writing tasks are often smaller blocks of cyan surrounded by brief periods of reference. The white-space in both graphs represents activities that aren't writing, revising, or using the primary software tools, but are no less important to the overall writing process. The block of white in the second half of the observed period was a longer break caused by a frustrating period of writer's block. Without this break, I probably wouldn't have been able to wrap up the paper. These white spaces are likely the most important part to look at in any future research. This observed period also only looked at a single device. Having spent a lot of time working in areas where mobile devices aren't allowed, my phone is not nearly as important to me as it might be to many of my peers and at no point during the observed period did I interact with it, even on breaks. However, this experience is not what I would expect to be the norm; nearly half of all worldwide internet usage was performed over mobile and tablet devices and is trending upwards ("Mobile Vs. Desktop Internet Usage (Latest 2020 Data)"). Future research in this way should have a way to account for time interacting with mobile devices, which I would currently be unable to track using the tools I had.

Sondra Perl asked back in 1979, "Can [unskilled college writers'] writing processes be analyzed in a systemic, replicable manner?" (Perl). Her conclusions answer yes, that the writing process can be coded in ways that deliver actionable conclusions in a replicable way. Modern technology has greatly magnified the scope and scale at which these behaviors can be captured. This study used a relatively simple method of coding – rewatching a recorded session and hand coding notes and actions into a spreadsheet. The process of conducting this initial self-study revealed many ideas for future research, such as implementing eye-tracking software or tools that track the active window or keystrokes. Even as we learn that the writing process is far from the structured and systematic process that it is often assumed, these technological advancements allow researchers to drill down further into the writing process and see smaller moments that build up to the total writing process.

It is important as writing instructors that we understand that there isn't a single strategy towards writing. Imparting a single "writing process" onto students fails to capture the nuanced and recursive reality of writing, that reality captured in one form in the data collected during this self-study. Instead of teaching students *the* writing process, we have to teach students how to unlock *their* writing process in the way most productive for them. Showing students how to analyze their writing will always be an important part of this process.

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